

What we know. What we assumed. What must be added.

A source-by-source review for the 1 km iLILA concept

01 THERMAL

02 OPTICAL

03 READOUT

04 ENVIRONMENT

05 TOTAL

Measurement definition and controlling requirement

Every curve must end in the same calibrated output before it can enter the budget

OUTPUT

Equivalent differential displacement
[m/√Hz]

BAND

1-100 mHz
(10^{-3} - 10^{-1} Hz)

REQUIREMENT

1×10^{-15} m/√Hz
preliminary

ARCHITECTURE

1 km equal-arm
dark-fringe interferometer

WHAT “FIXED MIRRORS” CHANGES

- No suspension thermal-noise model is imported from terrestrial LIGO.
- Brownian motion does not disappear: coatings, substrates, mounts, flexures and actuator interfaces still dissipate energy.
- Environmental motion reaches the readout through monuments, regolith, tilt, alignment and optical phase—not a single generic seismic curve.
- Dark-fringe common-mode rejection makes residual arm mismatch and contrast defect central coupling parameters.

MODEL GOVERNANCE RULE

A term is not “known” merely because a formula exists.

To close a term we need:

- Measured or specified input spectrum
- Validated transfer function to DARM
- Uncertainty or margin
- Frequency range and operating state

Noise-source status at a glance

Known physics is not the same as a closed iLILA allocation

1	Coating Brownian	Heritage proxy	No coating stack or first-principles curve
2	Substrate Brownian	Assumed curve	Material loss and beam radius unverified
3	Mount/actuator Brownian	Missing	Control residual exists; Brownian does not
4	Optical-bench thermal	Assumed curve	Temperature ASD + transfer function missing
5	Shot noise	Formula closed	Received power and losses assumed
6	Laser frequency/phase	Assumed curve	Residual spectrum and ΔL unmeasured
7	Laser RIN	Assumed curve	RIN and amplitude→phase coupling unmeasured
8	Pointing/wavefront	Assumed curve	Geometry and jitter transfer missing
9	Dust/scattered light	Heritage + assumed	Lunar BRDF, dust and motion model missing
10	Photodetector/electronics	Assumed aggregate	Component noise budget missing
11	Lunar seismic	Missing	LOVBB is a reference, not background
12	Quadrature total	Provisional	Incomplete inputs + no covariance

How the budget is combined—and where RSS fails

Use PSD/ASD consistently and retain correlations when they exist

ONE-SIDED ASD

$$A_i(f) = \sqrt{S_{x,i}(f)}$$

Units: m/√Hz. Plot and compare ASDs, but add independent noises in power.

UNCORRELATED TOTAL

$$A_{\text{tot}}(f) = \sqrt{\sum_i A_i^2(f)}$$

This is the current model rule. It is valid only when cross-spectra are negligible.

CORRELATED TOTAL

$$S_{\text{tot}} = \sum S_i + 2 \sum \text{Re}[S_{ij}]$$

Required for common laser noise, seismic coherence, control coupling and shared thermal paths.

ALLOCATION CHECK

If 11 sources shared the requirement equally:

$$10^{-15} / \sqrt{11}$$

$$\approx 3.0 \times 10^{-16} \text{ m}/\sqrt{\text{Hz}}$$

This is a planning allocation, not a physics prediction.

Keep margin for:

- Model uncertainty
- Correlations
- Transient states
- Calibration error

Coating Brownian noise

Equivalent differential displacement noise in the 1–100 mHz iLILA band

HERITAGE PROXY

WORKING EQUATION

$$S_x(f) = [2kBT/(\pi^2 f^2)] \cdot W_{\text{diss}}/F_0^2$$

Levin's direct fluctuation-dissipation method; coating stack and beam weighting determine W_{diss} .

CURRENT MODEL ASD

1 mHz

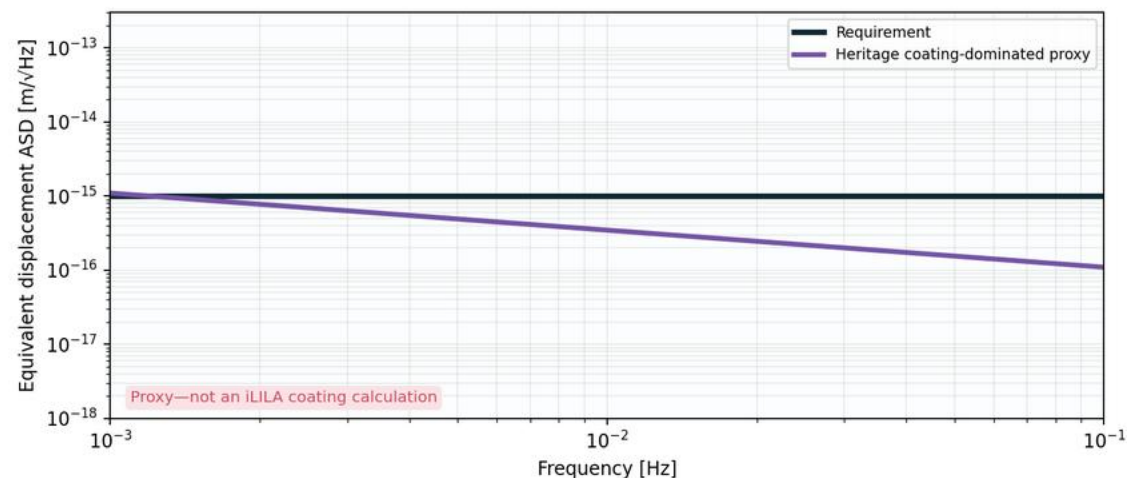
1.10e-15

10 mHz

3.46e-16

100 mHz

1.10e-16



WHAT WE KNOW

- Mechanical loss in dielectric layers produces surface displacement noise.
- Older LILA work provides a coating-dominated $f^{-1/2}$ ASD scale.
- Coating Brownian can dominate precision interferometers even when substrate loss is low.

CURRENT ASSUMPTION / PROXY

- The total currently uses the heritage thermal-target curve as a coating proxy.
- 300 K is inherited as a model control.
- No explicit coating geometry or material stack is represented.

WHAT MUST BE FIXED OR ADDED

- List every coated optic that couples to DARM, including local optics and reflector implementation.
- Specify layer materials, thicknesses, elastic properties and loss angles.
- Calculate finite-optic/CCR geometry with the actual beam radius and temperature.
- Replace the proxy and propagate parameter uncertainty.

Substrate Brownian noise

Equivalent differential displacement noise in the 1–100 mHz iLILA band

ASSUMED CURVE

WORKING EQUATION

$$S_x = 2kBT(1-\sigma^2)\phi / (\pi^{3/2} f wY)$$

Displayed quantity is $\sqrt{S_x}$. The current expression assumes a semi-infinite substrate and Gaussian pressure profile.

CURRENT MODEL ASD

1 mHz

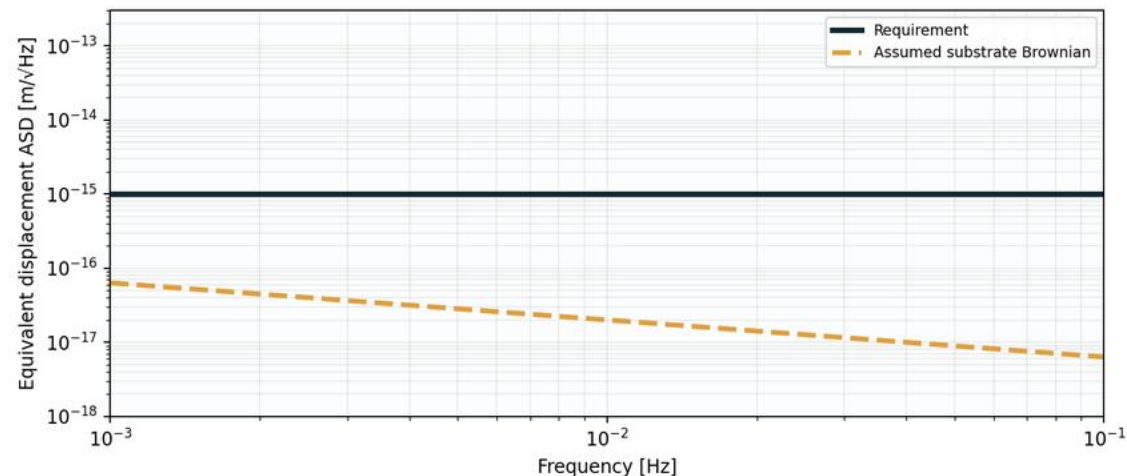
6.33e-17

10 mHz

2.00e-17

100 mHz

6.33e-18



WHAT WE KNOW

- A first-principles Brownian formula is implemented.
- Noise falls as $f^{-1/2}$ in ASD and increases with temperature and mechanical loss.
- Larger beam radius averages more independent surface motion.

CURRENT ASSUMPTION / PROXY

- Fused silica: $Y = 72$ GPa, $\sigma = 0.17$, $\phi = 10^{-7}$.
- Temperature = 300 K.
- Narrowest beam radius = 0.5 mm.

WHAT MUST BE FIXED OR ADDED

- Confirm actual substrate material, optic size and operating temperature.
- Measure or source $\phi(f,T)$ for the exact material and surface preparation.
- Use finite-size correction and the actual beam radius on each optic.
- Check whether CCR response can be represented as a conventional mirror substrate.

Mirror-mount and actuator Brownian noise

NOT MODELED

Equivalent differential displacement noise in the 1–100 mHz iLILA band

WORKING EQUATION

$$S_x(f) = [4kBT/(2\pi f)] \cdot \text{Im}[\chi(f)]$$

Generic fluctuation-dissipation form. $\chi(f)$ is the mechanical compliance from generalized force to optical displacement.

CURRENT MODEL ASD

1 mHz

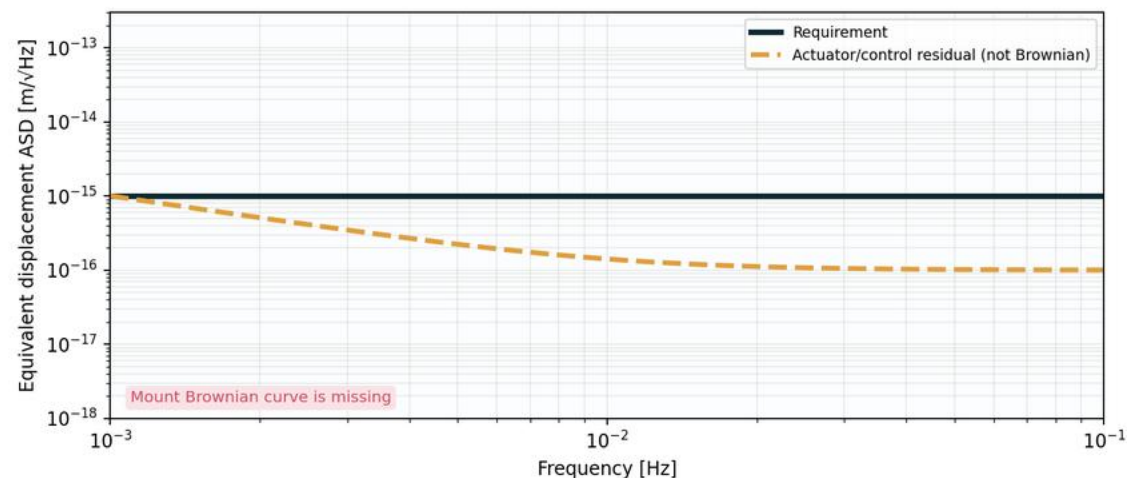
1.00e-15*

10 mHz

1.41e-16*

100 mHz

1.00e-16*



WHAT WE KNOW

- Fixed mirrors remain coupled to lossy mounts, adhesive joints, flexures, actuators and monuments.
- Thermal noise depends on the complete mechanical transfer function and loss distribution.
- Control-force residuals and Brownian motion are physically different terms.

CURRENT ASSUMPTION / PROXY

- A dashed actuator/control residual is present: 10^{-16} m/√Hz at 0.1 Hz with a 10 mHz corner.
- That curve is currently a placeholder for control performance only.
- It must not be relabeled as mount Brownian noise.

WHAT MUST BE FIXED OR ADDED

- Define mount, adhesive, preload, actuator and monument geometry.
- Measure modal frequencies, Q values and loss angles under lunar thermal/vacuum conditions.
- Compute compliance from each dissipative element to optical phase.
- Add Brownian and control residual as separate curves.

Optical-bench thermal noise

Equivalent differential displacement noise in the 1–100 mHz iLILA band

ASSUMED—DOMINANT LOW f

WORKING EQUATION

$$A_{\text{bench}}(f) = L_{\text{unbal}} \cdot \alpha_{\text{eff}} \cdot AT(f)$$

Simple thermoelastic path-length model; AT is temperature ASD, not the ± 10 mK stability bound.

CURRENT MODEL ASD

1 mHz

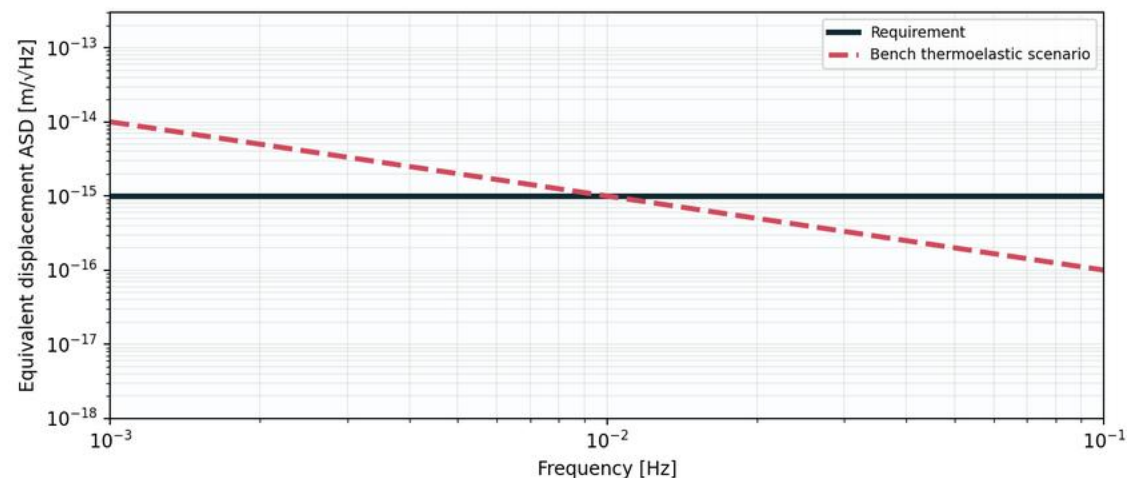
$1.00\text{e-}14$

10 mHz

$1.00\text{e-}15$

100 mHz

$1.00\text{e-}16$



WHAT WE KNOW

- The proposal specifies optical-bench stability of ± 10 mK.
- Thermal expansion couples through non-common path length and material CTE.
- Current model makes this the dominant term near 1–10 mHz.

CURRENT ASSUMPTION / PROXY

- $L_{\text{unbal}} = 0.1$ m; $\alpha_{\text{eff}} = 10^{-8}$ K⁻¹.
- $AT(0.1 \text{ Hz}) = 10^{-7}$ K/√Hz with a 1/f ASD.
- This produces 10^{-14} m/√Hz at 1 mHz.

WHAT MUST BE FIXED OR ADDED

- Convert the ± 10 mK bound into an in-band temperature ASD.
- Build a thermal network for bench, optics, mounts and enclosure.
- Measure temperature→DARM transfer functions and time delays.
- Separate expansion, thermo-refractive and alignment-mediated coupling.

Shot noise

FORMULA KNOWN—LINK ASSUMED

Equivalent differential displacement noise in the 1–100 mHz iLILA band

WORKING EQUATION

$$A_{\text{shot}} = \sqrt{[\hbar c \lambda / (2 \pi \eta P_{\text{det}})]}$$

Working Michelson phase-to-displacement approximation; the full readout topology may add optical gain factors.

CURRENT MODEL ASD

1 mHz

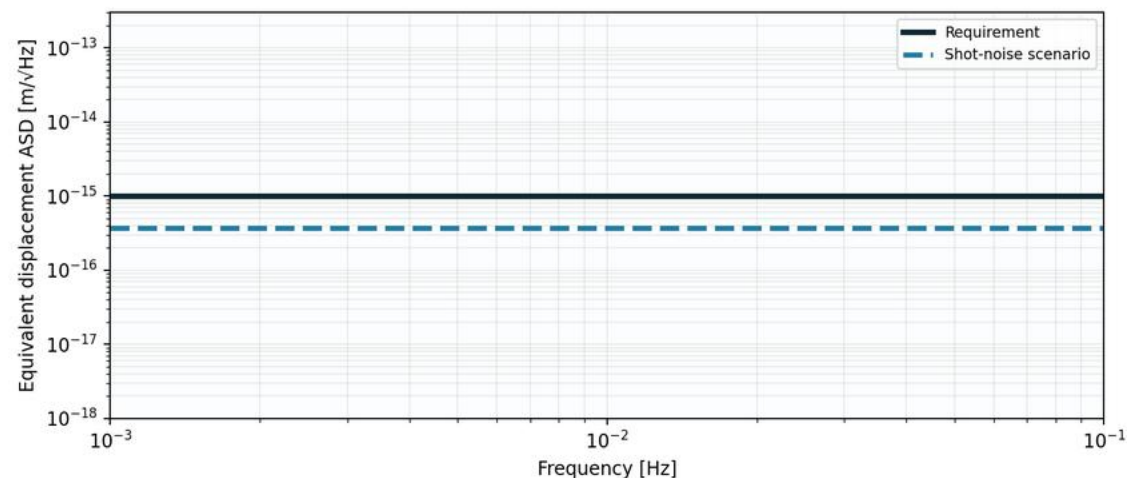
3.66e-16

10 mHz

3.66e-16

100 mHz

3.66e-16



WHAT WE KNOW

- $\lambda = 1064$ nm; proposal laser is 1 W input and 2 W after amplification.
- Shot ASD scales as $P_{\text{det}}^{-1/2}$.
- The modeled floor is white across 1–100 mHz.

CURRENT ASSUMPTION / PROXY

- 1 W launched per arm after splitting.
- 5% round-trip optical efficiency and 80% quantum efficiency.
- Effective detected power = 40 mW.

WHAT MUST BE FIXED OR ADDED

- Complete transmitter→CCR→receiver link budget.
- Measure received carrier power, contrast defect and local-oscillator power.
- Include photodiode saturation, mode mismatch and readout optical gain.
- Recalculate for realistic lunar dust, pointing and reflector losses.

Laser frequency and phase noise

ASSUMED CURVE

Equivalent differential displacement noise in the 1–100 mHz iLILA band

WORKING EQUATION

$$A\nu(f) = \Delta L \cdot A\delta\nu(f) / \nu_0$$

Equal arms cancel common laser phase to first order; macroscopic mismatch and control residual set the coupling.

CURRENT MODEL ASD

1 mHz

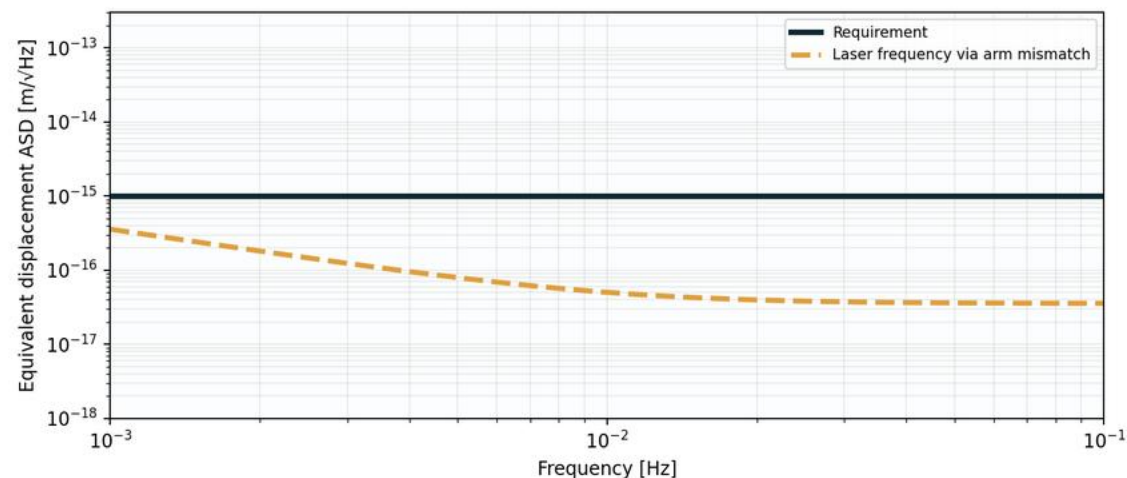
3.57e-16

10 mHz

5.02e-17

100 mHz

3.57e-17



WHAT WE KNOW

- $\nu_0 = c/\lambda$ for the 1064 nm laser.
- The proposal includes a reference cavity, stabilizer cavity and mode cleaners.
- Frequency coupling is proportional to residual arm mismatch, not the full 1 km arm.

CURRENT ASSUMPTION / PROXY

- $\Delta L = 1$ cm.
- Stabilized $A\delta\nu = 1$ Hz/√Hz at high frequency with a 10 mHz corner.
- The spectral shape is $A\nu[1+(f_c/f)^2]$.

WHAT MUST BE FIXED OR ADDED

- Measure free-running and closed-loop frequency ASD down to 1 mHz.
- Define arm-matching tolerance and drift budget.
- Model reference-cavity thermal and seismic noise.
- Include contrast defect and frequency→amplitude/readout cross-couplings.

Laser relative-intensity noise

Equivalent differential displacement noise in the 1–100 mHz iLILA band

ASSUMED CURVE

WORKING EQUATION

$$\text{ARIN}(f) = [\lambda/(4\pi)] \cdot \text{CA} \rightarrow \phi \cdot \text{RIN}(f)$$

RIN matters only through an amplitude-to-phase/readout coupling; an ideal dark fringe rejects common intensity noise.

CURRENT MODEL ASD

1 mHz

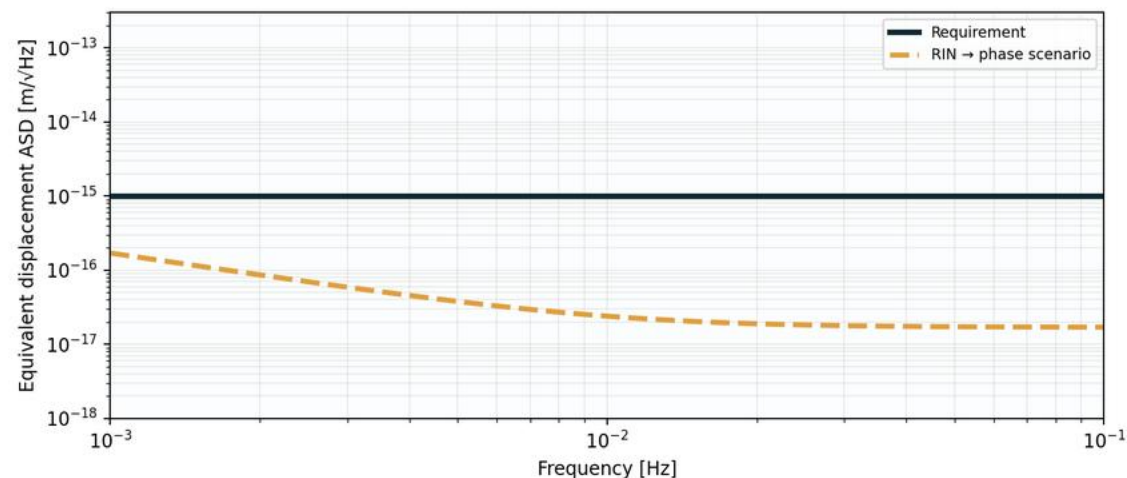
$1.70\text{e-}16$

10 mHz

$2.39\text{e-}17$

100 mHz

$1.70\text{e-}17$



WHAT WE KNOW

- The proposal includes laser conditioning and dark-fringe recombination.
- RIN can couple through contrast defect, detector imbalance, radiation pressure and electronics nonlinearity.
- The current model represents only a phase-equivalent path.

CURRENT ASSUMPTION / PROXY

- $\text{RIN}(0.1 \text{ Hz}) = 2 \times 10^{-6}/\sqrt{\text{Hz}}$.
- Corner frequency = 10 mHz.
- $\text{CA} \rightarrow \phi = 10^{-4}$ rad per unit RIN.

WHAT MUST BE FIXED OR ADDED

- Measure RIN before/after stabilization in the target band.
- Measure dark-port RIN→DARM transfer versus contrast defect and alignment.
- Separate direct photodetector amplitude noise from true optical phase coupling.
- Include radiation-pressure coupling if relevant for the fixed optic/mount system.

Beam pointing and wavefront coupling

ASSUMED—IMPORTANT LOW f

Equivalent differential displacement noise in the 1–100 mHz iLILA band

WORKING EQUATION

$$A_{\text{point}}(f) \approx 2 \text{ doffset} \cdot A\theta(f)$$

First-order geometric proxy. Real coupling depends on telescope propagation, wavefront error and CCR return geometry.

CURRENT MODEL ASD

1 mHz

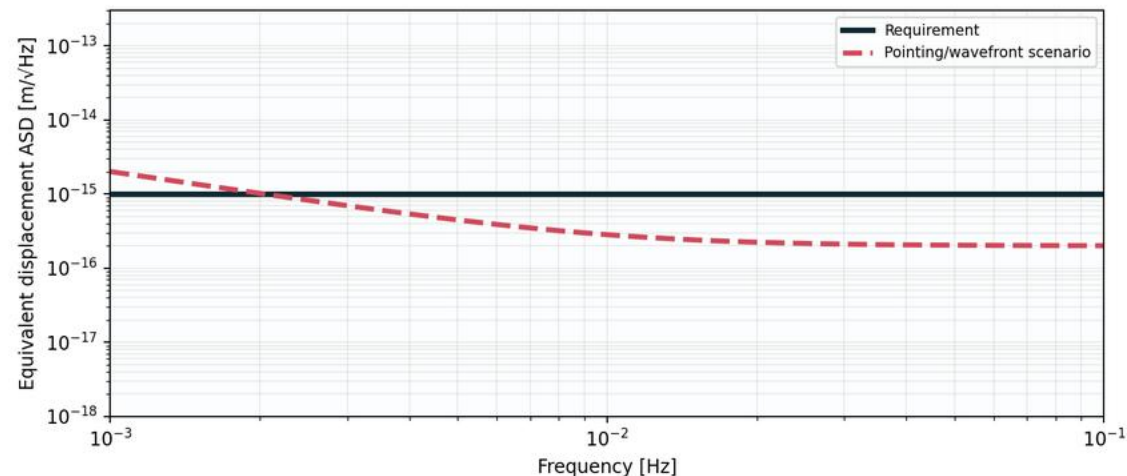
2.01e-15

10 mHz

2.83e-16

100 mHz

2.01e-16



WHAT WE KNOW

- The proposal requires $\approx 3^\circ$ far-reflector alignment and uses pointing/control hardware.
- Angular motion becomes path-length noise when the beam is off-center or wavefronts are mismatched.
- Current curve is the second-largest term at 1 mHz.

CURRENT ASSUMPTION / PROXY

- Beam-spot offset = 100 μm .
- Angular jitter = 10^{-12} rad/ $\sqrt{\text{Hz}}$ at 0.1 Hz with a 10 mHz corner.
- A factor of two is used for reflection geometry.

WHAT MUST BE FIXED OR ADDED

- Propagate measured angular jitter through the full transmitter/receiver optics.
- Compute beam walk, wavefront curvature and CCR angular response.
- Measure spot offsets and alignment-control residuals.
- Add temperature-driven pointing and higher-order-mode coupling.

Dust and scattered-light noise

HERITAGE + ASSUMED

Equivalent differential displacement noise in the 1–100 mHz iLILA band

WORKING EQUATION

$$\phi_{sc}(t) \approx (4\pi/\lambda) x_{sc}(t); \quad A_x \approx (\lambda/4\pi) A_\phi$$

Backscattered fields acquire phase from moving surfaces and interfere with the main field; dust also changes loss and scattering.

CURRENT MODEL ASD

dust

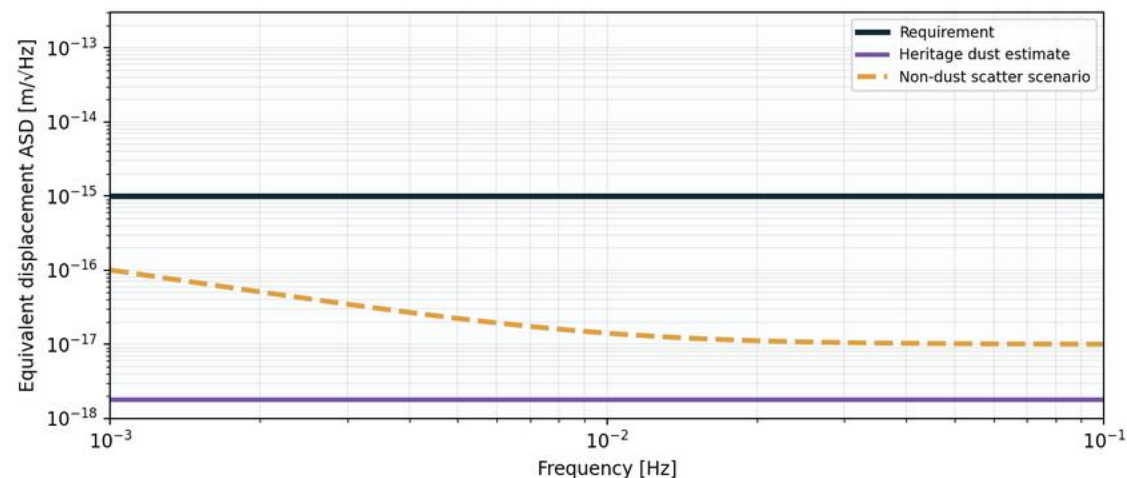
1.79e-18

scatter@10mHz

1.41e-17

scatter@1mHz

1.00e-16



WHAT WE KNOW

- The proposal identifies lunar dust and optical scatter as environmental risks.
- Older LILA work supplies a dust PSD scale that can be rescaled with density, radius, speed and beam size.
- LIGO experience shows scattered light is difficult to predict a priori.

CURRENT ASSUMPTION / PROXY

- Dust: $n = 3 \times 10^{-3} \text{ m}^{-3}$, radius 0.7 μm , speed 300 m/s, beam radius 1 cm.
- Heritage dust ASD is $1.79 \times 10^{-18} \text{ m}/\sqrt{\text{Hz}}$.
- Non-dust scatter = $10^{-17} \text{ m}/\sqrt{\text{Hz}}$ at 0.1 Hz with 10 mHz corner.

WHAT MUST BE FIXED OR ADDED

- Measure lunar dust density/size/velocity distributions by site and illumination.
- Build stray-light inventory: BRDF, surfaces, path lengths and motion spectra.
- Include CCR contamination, contrast loss and received-power degradation.
- Validate nonlinear/upconverted scatter with injection tests.

Photodetector and electronics noise

ASSUMED AGGREGATE

Equivalent differential displacement noise in the 1–100 mHz iLILA band

WORKING EQUATION

$$A_{\text{read}} = [\lambda/(4\pi)] \cdot A\phi_{\text{read}}$$

Current model converts one combined phase floor to displacement; it does not expose component-level limits.

CURRENT MODEL ASD

1 mHz

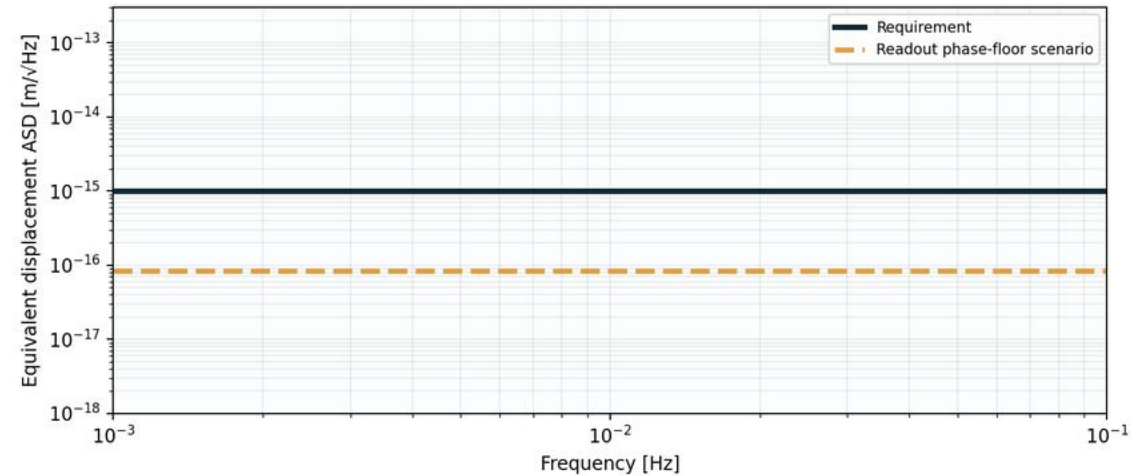
8.47e-17

10 mHz

8.47e-17

100 mHz

8.47e-17



WHAT WE KNOW

- The proposal includes photodetectors, frequency/control electronics and digitization.
- Readout noise can include dark current, shot noise, TIA voltage/current noise, ADC noise, clock jitter and demodulation phase noise.
- Some terms scale with optical power; others do not.

CURRENT ASSUMPTION / PROXY

- Combined phase floor = 10^{-9} rad/√Hz.
- Equivalent displacement = 8.47×10^{-17} m/√Hz, white.
- This is separate from the optical shot-noise curve.

WHAT MUST BE FIXED OR ADDED

- Build component noise model in A/√Hz and V/√Hz before converting through optical gain.
- Specify photodiode responsivity, dark current, capacitance and saturation.
- Specify TIA topology, ADC resolution, sample clock and demodulation chain.
- Measure dark, illuminated and closed-loop spectra.

Lunar seismic scenarios

Equivalent differential displacement noise in the 1–100 mHz iLILA band

BACKGROUND NOT MODELED

WORKING EQUATION

$$Ax,abs(f) = Aa(f)/(2\pi f)^2; ADARM = |Hdiff(f)| Ax$$

Double integration gives absolute motion only.
Differential interferometer coupling requires coherence and a site/structure transfer function.

CURRENT MODEL ASD

LOVBB ref

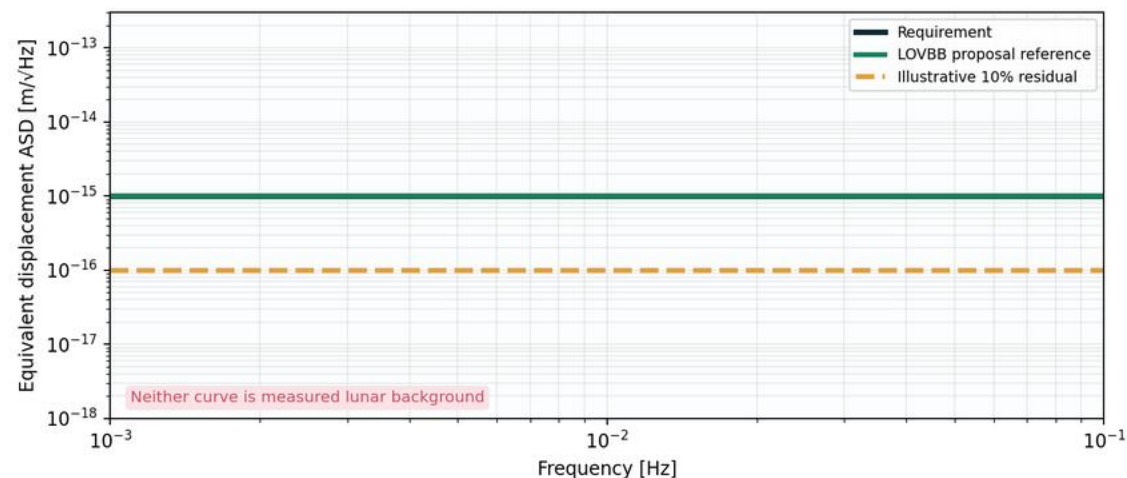
1.00e-15

10× residual

1.00e-16

background

MISSING



WHAT WE KNOW

- The proposal targets LOVBB self-noise of 10^{-11} m/s²/√Hz and quotes $\approx 10^{-15}$ m/√Hz equivalent strainmeter displacement.
- Apollo recorded deep/shallow moonquakes, thermal events and impacts with long lunar codas.
- The target band overlaps Apollo long-period coverage but requires modern reprocessing and site translation.

CURRENT ASSUMPTION / PROXY

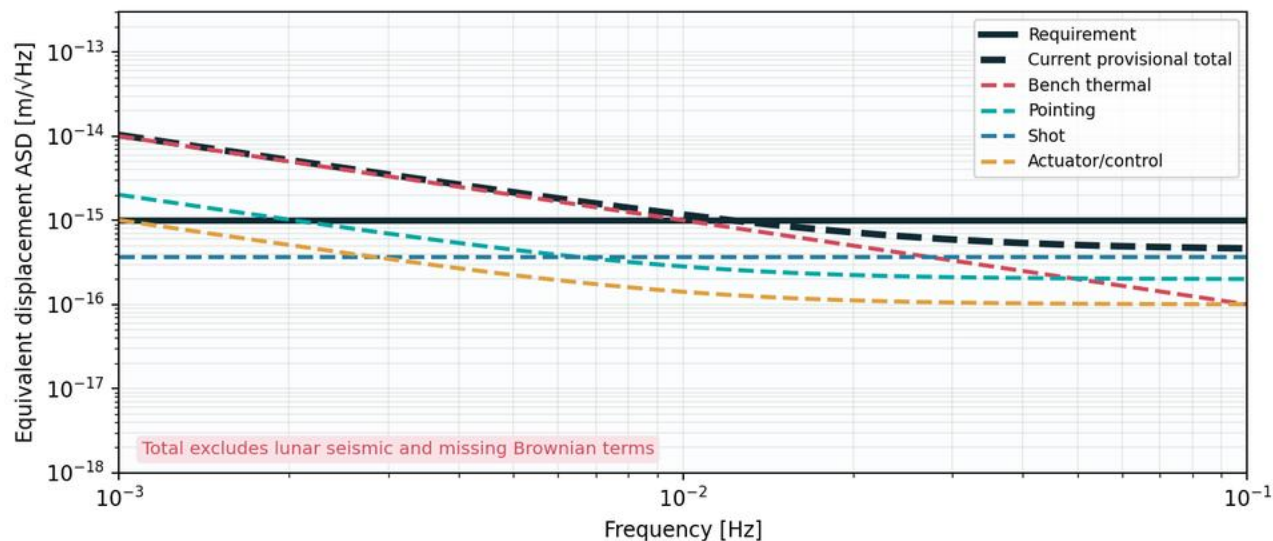
- LOVBB equivalent = 10^{-15} m/√Hz in band.
- Illustrative residual after 10× subtraction = 10^{-16} m/√Hz.
- Neither is interpreted as actual lunar ground motion.

WHAT MUST BE FIXED OR ADDED

- Create quiet-time and event-conditioned lunar acceleration ASDs from Apollo/new models.
- Estimate 1 km spatial coherence and two-arm common-mode rejection.
- Model regolith→foundation→bench and tilt→length transfer functions.
- Demonstrate subtraction with realistic sensor noise, geometry, causality and transients.

Quadrature-summed total: useful, but not closed

The current total answers “what do our placeholders imply?”—not “what will iLILA achieve?”



CURRENT RESULT

1 mHz 1.03×10^{-14}

dominant: bench thermal

10 mHz 1.17×10^{-15}

dominant: bench thermal

100 mHz 4.64×10^{-16}

dominant: shot noise

m/√Hz

INCLUDED

- Heritage coating proxy
- Substrate, shot, frequency, RIN
- Pointing, bench, readout
- Control residual + scatter

EXCLUDED / INCOMPLETE

- First-principles coating
- Mount/actuator Brownian
- Lunar background + coupling
- Correlations and uncertainty

STRUCTURAL FIX

- Separate proxies from allocations
- Add covariance where needed
- Carry model uncertainty
- Define observing-state totals

Complete numeric input ledger — part 1

P = proposal value • H = heritage input • A = editable assumption

PARAMETER	VALUE	CLASS
arm length (m)	1000	P
laboratory arm length (m)	1	P
opening angle (deg)	90	P
wavelength (m)	1.1e-6	P
input laser power (W)	1	P
amplified laser power (W)	2	P
target frequency min (Hz)	1e-3	P
target frequency max (Hz)	0.1	P
displacement requirement m/√Hz	1e-15	P
optical bench temperature stability (K)	0.01	P
lovbb acceleration asd m/s ² /√Hz	1e-11	P
lovbb equivalent displacement m/√Hz	1e-15	P
far reflector diameter (m)	0.1	P
far reflector alignment tolerance (deg)	3	P

PARAMETER	VALUE	CLASS
temperature (K)	300	A
assumed power per arm (W)	1	A
assumed round trip optical efficiency	0.05	A
assumed photodiode quantum efficiency	0.8	A
heritage thermal target psd 1Hz m ² /Hz	1.2e-33	H
heritage thermal conservative psd 1Hz m ² /Hz	6e-32	H
heritage dust psd reference m ² /Hz	1.6e-35	H
dust density m ⁻³	3e-3	A
dust radius (m)	7e-7	A
dust speed m per s	300	A
assumed arm beam radius (m)	0.01	A
assumed arm length mismatch (m)	0.01	A
assumed laser frequency asd 1Hz Hz/√Hz	1	A
laser frequency corner (Hz)	0.01	A

Complete numeric input ledger — part 2

Every A-class input needs a measurement, specification or justified margin

PARAMETER	VALUE	CLASS
substrate poisson ratio	0.17	A
substrate youngs modulus (Pa)	7.2e+10	A
substrate loss angle	1e-7	A
assumed narrowest beam radius (m)	5e-4	A
assumed rin asd 0p1Hz/√Hz	2e-6	A
rin low frequency corner (Hz)	0.01	A
assumed amplitude to phase rad per rin	1e-4	A
assumed beam spot offset (m)	1e-4	A
assumed angular jitter asd 0p1Hz rad/√Hz	1e-12	A
angular jitter corner (Hz)	0.01	A
assumed local unbalanced path (m)	0.1	A
assumed bench effective cte /K	1e-8	A
assumed temperature asd 0p1Hz K per rHz	1e-7	A

PARAMETER	VALUE	CLASS
temperature asd power	1	A
assumed readout phase floor rad/√Hz	1e-9	A
assumed actuator residual asd 0p1Hz m/√Hz	1e-16	A
actuator residual corner (Hz)	0.01	A
assumed scattered light asd 0p1Hz m/√Hz	1e-17	A
scattered light corner (Hz)	0.01	A
apollo instrument floor m/s ² /√Hz	1e-10	A
apollo diagnostic min (Hz)	0.05	A
apollo diagnostic max (Hz)	1	A
assumed seismic subtraction residual fraction	0.1	A
frequency min (Hz)	1e-4	A
frequency max (Hz)	1	A
frequency points	1601	A

Noise-budget closure plan

Convert the working model into a traceable requirement, measurement and uncertainty system

A**DEFINE**

- Freeze optical topology and readout gain
- Name every DARM-coupled optic/mount/path
- Define nominal, survival and observing states

B**MEASURE**

- Laser frequency/RIN/pointing spectra
- Received power and electronics floors
- Temperature and mechanical transfer functions

C**MODEL**

- Coating/substrate/mount FDT models
- Optical propagation + scattered light
- Lunar seismic coherence and coupling

D**VALIDATE**

- 1 m demonstrator noise injections
- Transfer-function closure tests
- Uncertainty and margin review

E**ALLOCATE**

- Component requirements by frequency
- RSS/covariance total and margins
- Ownership, acceptance test and update cadence

Immediate priorities: lunar seismic model • bench temperature ASD/transfer • received-power budget • mount Brownian model

REFERENCES

Technical references and controlling inputs

Primary sources used to define mechanisms, equations and open questions

LILA CONCEPT	LILA: Listening for the First Cosmic Sound — Templeton proposal, August 2026.
MECHANICAL THERMAL NOISE	P. R. Saulson, Phys. Rev. D 42, 2437 (1990). doi:10.1103/PhysRevD.42.2437
TEST-MASS FDT	Y. Levin, Phys. Rev. D 57, 659 (1998). doi:10.1103/PhysRevD.57.659
COATING BROWNIAN	G. M. Harry et al., Class. Quantum Grav. 19, 897 (2002). arXiv:gr-qc/0109073
SHOT NOISE	T. T. Lyons, M. W. Regehr & F. J. Raab, Applied Optics 39, 6761 (2000).
FREQUENCY NOISE	Initial LIGO Prestabilized Laser Design Requirements, LIGO-T950030.
ALIGNMENT COUPLING	P. Fritschel et al., Applied Optics 37, 6734 (1998).
SCATTERED LIGHT	Advanced LIGO Stray Light Control Design Requirements, LIGO-T070061.
LUNAR SEISMIC	Y. Nakamura et al., Apollo Lunar Seismic Experiment—Final Summary (1982).
WORKING ARTIFACTS	LILA V2 Python model, full spectral CSV, reference table and assumption register.
Scope note: formulas are working reductions; the final budget must use the actual optical topology, geometry and calibrated transfer functions.	